Hermetic-Sheath Toroidal Pickup: BEM Calculation Log

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May 11, 2026

Abstract

Running log of the unit-current boundary-element calculation for the hermetic Meissner-sheath toroidal axion pickup, following the v2 task specification in `cc_task_spec.md`. We extract the dimensionless coefficients $F(\beta;\alpha) \equiv V_{\rm eff}/R^3$ and $\ell(\beta;\alpha) \equiv L_p/(\mu_0 R)$ in the thin-annulus limit, test whether the favorable $Q \propto R_t^4$ scaling at fixed magnet cost survives parasitic inductance from slit and external return, and compute the crossover radius R_t^* at which the toroid figure of merit equals the long-solenoid Core baseline.

Contents

1 Setup and goal

We follow the notation of *Hermetic-Sheath Toroidal Axion Pickup* (Safdi 2026, included as `toroid_optimization.pdf` in this repository). All lengths are in units of the major radius R_t; $\mu_0 \equiv 1$; unit current $I \equiv 1$.

The magnet bore is

$$V_{\rm mag} = \{1 < s < 1+\beta,\ 0 < z < \alpha,\ 0 < \phi < 2\pi\}, \tag{1}$$

with DC field $\vec{B}_0 = B_{\rm max}(R/s)\hat{\phi}$.

Two scalar coefficients are extracted by polynomial fits in β at each α:

$$F(\beta) = f_1\beta + f_2\beta^2 + f_3\beta^3, \qquad \ell(\beta) = \ell_0 + \ell_1\beta + \ell_2\beta^2. \tag{2}$$

The decision criterion is in Sec. 1 of `cc_task_spec.md`: favorable scaling requires both f_1 statistically non-zero (3σ bootstrap) and ℓ_0 consistent with zero ($\mathrm{RSS}_{\ell_0=0} \leq 2\,\mathrm{RSS}_{\mathrm{free}}$).

2 Geometry and topology

2.1 Sheath surface

Four pieces, joined at edges:

- Inner cylinder $\mathcal{S}_A$: $s = 1$, $z \in [0, \alpha]$.
- Top cap $\mathcal{S}_B$: $z = \alpha$, $s \in [1, 1 + \beta]$.
- Outer cylinder $\mathcal{S}_C$: $s = 1 + \beta$, $z \in [0, \alpha]$.
- Bottom cap $\mathcal{S}_D$: $z = 0$, $s \in [1, 1 + \beta]$.

Without the slit the sheath is topologically a torus T^2. Removing the finite-width slit ribbon (running once around the meridional rectangle $A \to B \to C \to D \to A$ at azimuthal half-width $w/2$ around $\phi = 0$) makes the sheath an annular cylinder. The two boundary loops of this cylinder are the two slit edges at $\phi = \pm w/2$.

2.2 Stream function and slit jump

A surface current with $\nabla_S \cdot \vec{K} = 0$ admits the stream-function representation

$$\vec{K} = \hat{n} \times \nabla_S \psi. \tag{3}$$

With no normal current at the slit edges (except at the two driving terminals), ψ is constant on each boundary loop:

$$\psi\big|_{\phi=+w/2} = 1, \qquad \psi\big|_{\phi=-w/2} = 0. \tag{4}$$

Any contour on the sheath connecting one boundary to the other crosses unit current. *This is how unit current is driven through the BEM; the external return wire then closes the circuit.*

2.3 External return

Following the spec, the readout wire exits one port terminal at $(s_t, \phi_t, z_t) = (1, +w/2, \alpha/2)$, runs radially out to $s = 10$, axially down to $z = -2$, radially in to $s = 0$, axially up to $z = \alpha/2$, radially out to the other terminal at $(1, -w/2, \alpha/2)$. Wire radius $r_{\mathrm{wire}} = 0.01$ (regularizer for self-energy). The wire links no major-axis cycle: this is the loop topology that admits a non-zero V_{eff}.

3 Numerical method

Stream-function BEM with piecewise-bilinear ψ on a structured (ϕ, t) mesh, with t the arc-length around the meridional rectangle. Adaptive azimuthal mesh refinement near the slit ($|\phi| < 3w/2$: element size $\leq w/3$; $3w/2 < |\phi| < 6w/2$: $\leq w$; remainder: ≤ 0.1 rad). Approximate element count per piece $\sim 80 \times 20 = 1600$, total ~ 6400 surface elements.

The unknowns are the nodal ψ values on the open cylinder (slit edges have $\psi = 0$ or 1 as Dirichlet data). The BEM matrix $\mathsf{A}_{ij} = \hat{n}_i \cdot \vec{B}(\vec{r}_i; \psi_j)$ is evaluated at element centroids, with 4-point Gauss quadrature on non-adjacent panels and analytical near-singular subtraction on adjacent panels. The external return wire contributes an additional incident field $\vec{B}_{\mathrm{wire}}$ on the RHS.

Outputs of solver. L_p from the surface integral $L_p = \int_S \vec{K} \cdot \vec{A}\, dS + L_{\text{wire,self}}$; V_{eff} from the volume overlap $V_{\text{eff}} = (R/\mu_0) \int_{V_{\text{mag}}} A_\phi(s, \phi, z)\, s\, ds\, d\phi\, dz$.

4 Cross-checks

TODO: Run XC2 (zero-slit limit), XC3 (reciprocity), XC4 (gauge invariance) at $\alpha = \beta = 1$ before production sweep.

5 Production sweep

TODO: Sweep $\alpha \in \{0.5, 1.0, 2.0\} \times \beta \in \{0.20, 0.12, 0.08, 0.05, 0.035, 0.025\}$ on lr7/lr8 SLURM partitions.

6 Fits and decision

TODO: Unconstrained and $\ell_0 = 0$-constrained nonlinear least-squares; bootstrap residuals to estimate uncertainties; apply decision table from spec Sec. 1.

7 Crossover radius R_t^*

TODO: If favorable: R_t^* at $C_{\text{tor}} \in \{0.1, 1, 10\}\,\text{m}^3 \cdot B_{\max}^2$. Flag $R_t^* > 1\,\text{m}$ as impractical.

8 Decision

TODO: Binary decision per α.

A Adversarial review log

This appendix records adversarial reviews and the responses to each.